

Lecture 23/Module 3

Sinusoidal Steady State Analysis

08/09/2026

Dr. Sudipta Kumar Sarkar

Assistant Professor (Research), SENSE



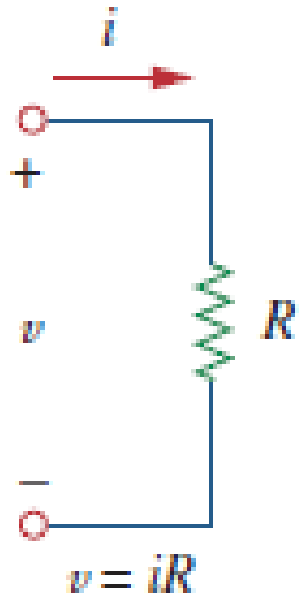
School of Electronics Engineering (SENSE)
VIT, Vellore, India

Today's Topics

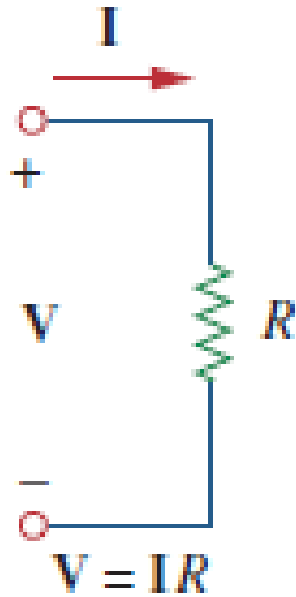
- Phasor Diagram of Circuit Elements
- Impedance & Admittance
- KCL & KVL in Frequency Domain

Phasor relationship for circuit elements

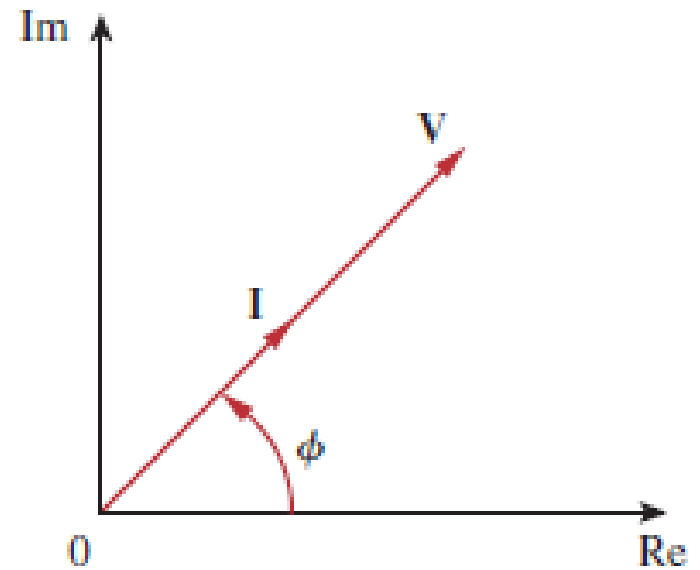
RESISTOR



(a)



(b)



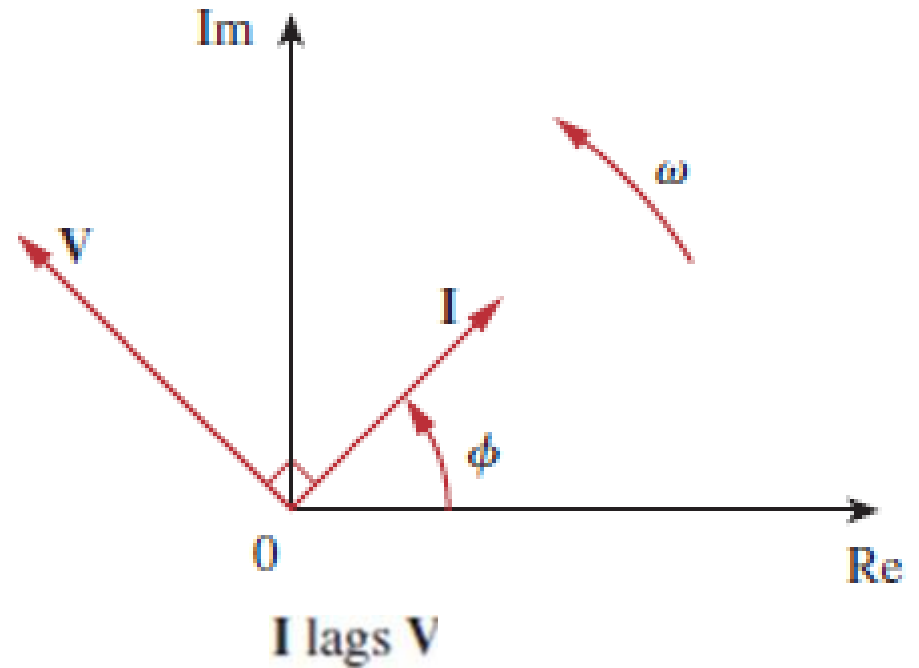
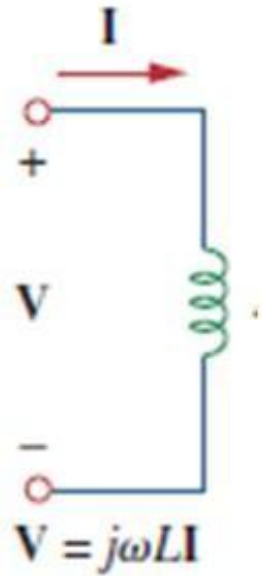
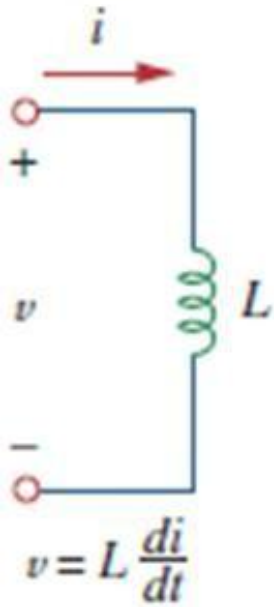
$$v = iR = RI_m \cos(\omega t + \phi)$$

$$V = RI_m \angle \phi$$

$$V = RI$$

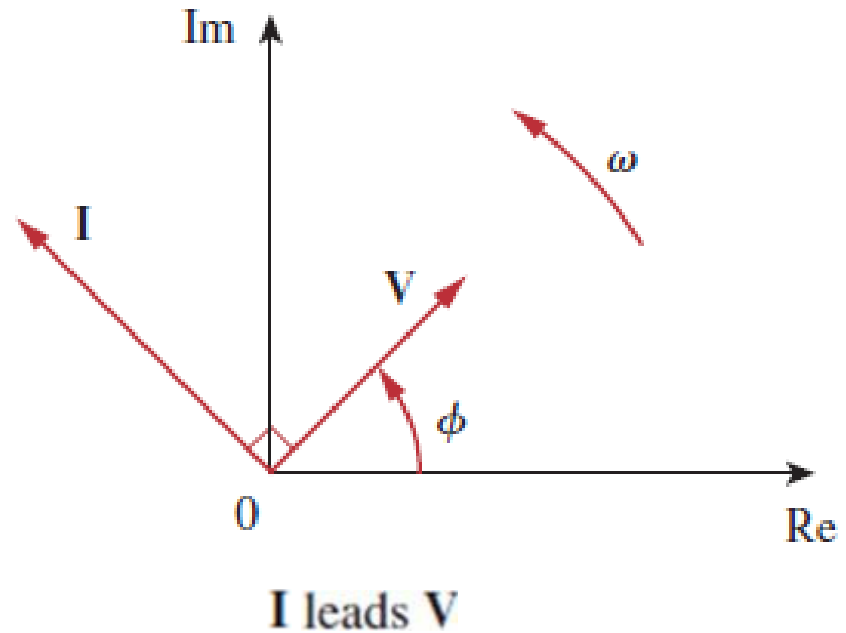
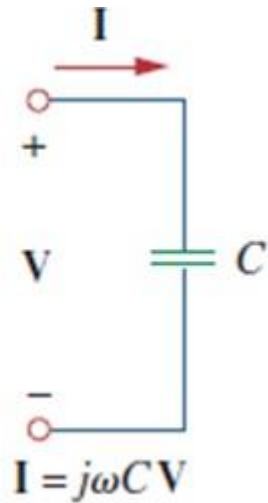
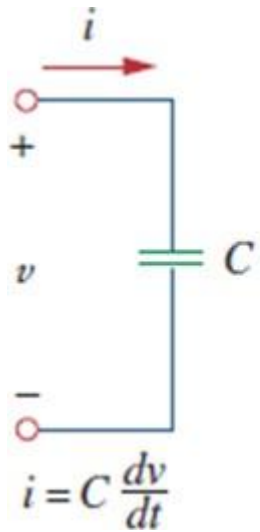
Phasor relationship for circuit elements

INDUCTOR



Phasor relationship for circuit elements

CAPACITOR



Phasor relationship for circuit elements

Voltage – Current relationship in circuit elements

Element	Time domain	Frequency domain
R	$v = Ri$	$V = RI$
L	$v = L \frac{di}{dt}$	$V = j\omega LI$
C	$i = C \frac{dv}{dt}$	$V = \frac{I}{j\omega C}$

Impedance

The impedance Z of a circuit is the ratio of the phasor voltage V to the phasor current I , measured in ohms.

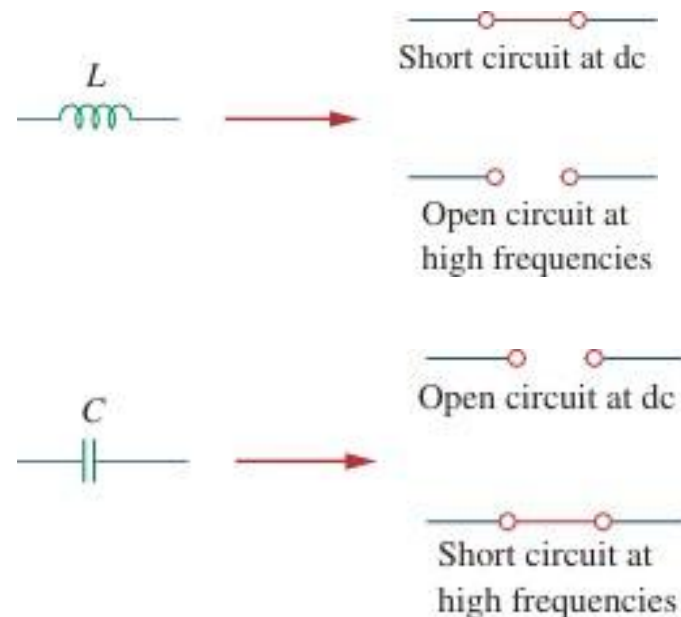
$$V = RI, \quad V = j\omega LI, \quad V = \frac{I}{j\omega C}$$

$$\frac{V}{I} = R, \quad \frac{V}{I} = j\omega L, \quad \frac{V}{I} = \frac{1}{j\omega C}$$

$$Z = \frac{V}{I} \quad \text{or} \quad V = ZI$$

Z is a frequency-dependent quantity known as impedance, measured in ohms.

Element	Impedance	Admittance
R	$Z = R$	$Y = \frac{1}{R}$
L	$Z = j\omega L$	$Y = \frac{1}{j\omega L}$
C	$Z = \frac{1}{j\omega C}$	$Y = j\omega C$



Phasor relationship for circuit elements

As a complex quantity, the impedance may be expressed in rectangular form as

$$\mathbf{Z} = R + jX$$

The impedance may also be expressed in polar form as

$$\mathbf{Z} = |\mathbf{Z}| \angle \theta$$

$$\mathbf{Z} = R + jX = |\mathbf{Z}| \angle \theta$$

$$|\mathbf{Z}| = \sqrt{R^2 + X^2}, \quad \theta = \tan^{-1} \frac{X}{R}$$

$$R = |\mathbf{Z}| \cos \theta, \quad X = |\mathbf{Z}| \sin \theta$$

Admittance

The admittance Y is the reciprocal of impedance, measured in Siemens (S).

The admittance Y of an element (or a circuit) is the ratio of the phasor current through it to the phasor voltage across it.

$$Y = \frac{1}{Z} = \frac{I}{V}$$

$$Y = G + jB$$

$$G + jB = \frac{1}{R + jX}$$

$$G + jB = \frac{1}{R + jX} \cdot \frac{R - jX}{R - jX} = \frac{R - jX}{R^2 + X^2}$$

$$G = \frac{R}{R^2 + X^2}, \quad B = -\frac{X}{R^2 + X^2}$$

Basic Laws and Analysis

Kirchoff's Laws in the Frequency domain

- KVL:

Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around a closed path (or loop) is zero.

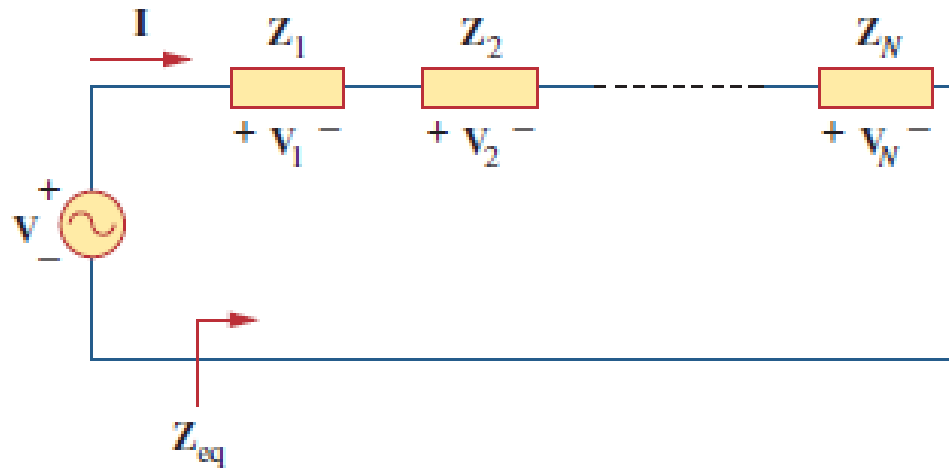
$$\mathbf{V}_1 + \mathbf{V}_2 + \cdots + \mathbf{V}_n = 0$$

- KCL:

Kirchhoff's current law (KCL) states that the algebraic sum of currents entering a node (or a closed boundary) is zero.

$$\mathbf{I}_1 + \mathbf{I}_2 + \cdots + \mathbf{I}_n = 0$$

N Impedances in Series

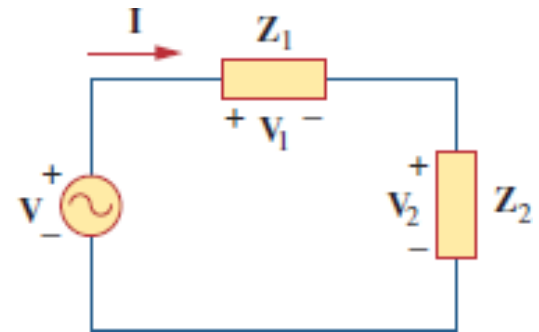


$$Z_{eq} = \frac{V}{I} = Z_1 + Z_2 + \cdots + Z_N$$

Voltage Division

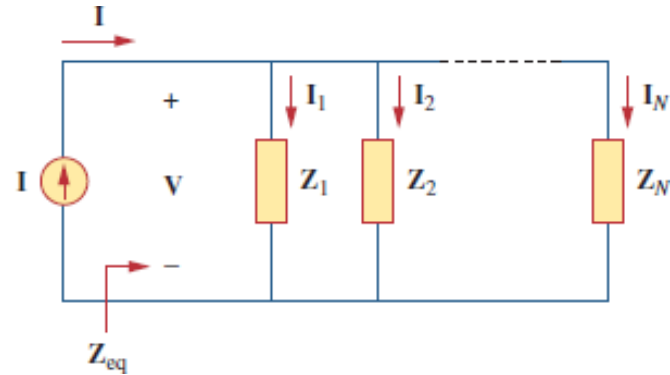
$$\mathbf{I} = \frac{\mathbf{V}}{\mathbf{Z}_1 + \mathbf{Z}_2}$$

$$\mathbf{V}_1 = \mathbf{Z}_1 \mathbf{I} \text{ and } \mathbf{V}_2 = \mathbf{Z}_2 \mathbf{I}$$



$$\mathbf{V}_1 = \frac{\mathbf{Z}_1}{\mathbf{Z}_1 + \mathbf{Z}_2} \mathbf{V}, \quad \mathbf{V}_2 = \frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2} \mathbf{V}$$

N Impedances in Parallel



$$I = I_1 + I_2 + \dots + I_N = V \left(\frac{1}{Z_1} + \frac{1}{Z_2} + \dots + \frac{1}{Z_N} \right)$$

$$\frac{1}{Z_{eq}} = \frac{I}{V} = \frac{1}{Z_1} + \frac{1}{Z_2} + \dots + \frac{1}{Z_N}$$

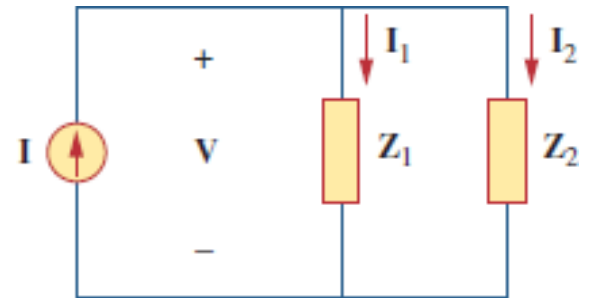
$$Y_{eq} = Y_1 + Y_2 + \dots + Y_N$$

Current Division

$$Z_{\text{eq}} = \frac{1}{Y_{\text{eq}}} = \frac{1}{Y_1 + Y_2} = \frac{1}{1/Z_1 + 1/Z_2} = \frac{Z_1 Z_2}{Z_1 + Z_2}$$

$$V = I Z_{\text{eq}} = I_1 Z_1 = I_2 Z_2$$

$$I_1 = \frac{Z_2}{Z_1 + Z_2} I, \quad I_2 = \frac{Z_1}{Z_1 + Z_2} I$$

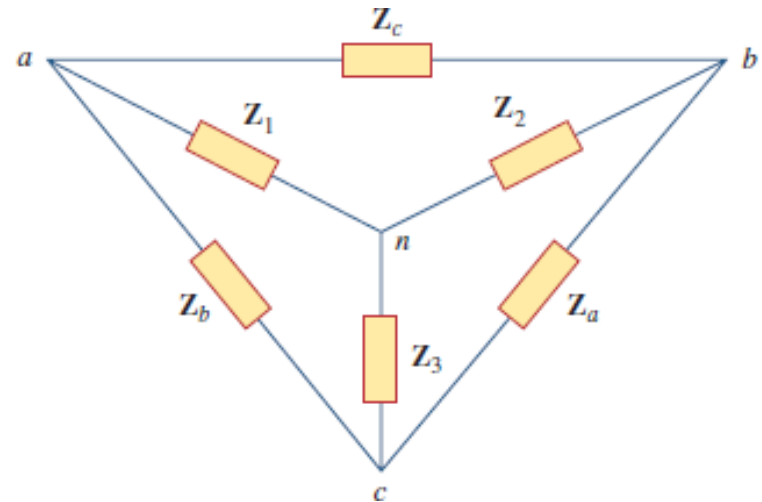


Wye to Delta conversion

$$Z_a = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_1}$$

$$Z_b = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_2}$$

$$Z_c = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_3}$$

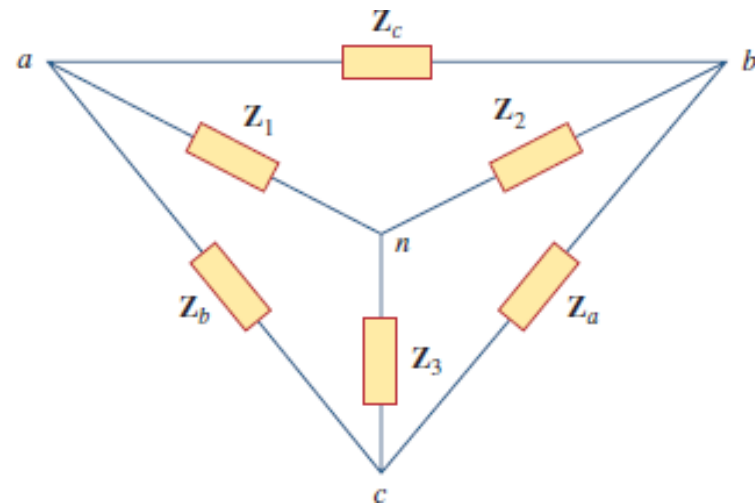


Delta to Wye conversion

$$Z_1 = \frac{Z_b Z_c}{Z_a + Z_b + Z_c}$$

$$Z_2 = \frac{Z_c Z_a}{Z_a + Z_b + Z_c}$$

$$Z_3 = \frac{Z_a Z_b}{Z_a + Z_b + Z_c}$$



A delta or wye circuit is said to be **balanced** if it has equal impedances in all three branches.

$$Z_{\Delta} = 3Z_Y \quad \text{or} \quad Z_Y = \frac{1}{3} Z_{\Delta}$$

where $Z_Y = Z_1 = Z_2 = Z_3$ and $Z_{\Delta} = Z_a = Z_b = Z_c$.

Exercise -1

Find $v(t)$ and $i(t)$ in the circuit shown

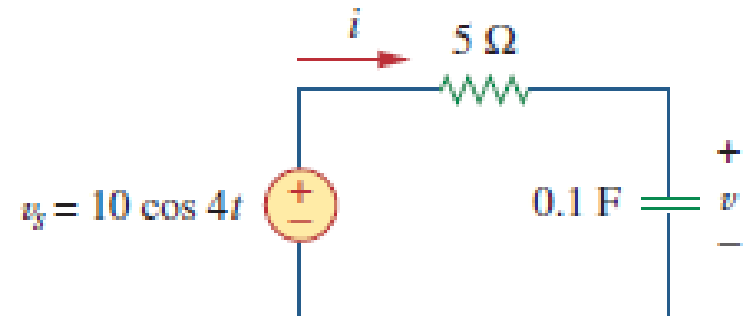
Solution:

From the voltage source $10 \cos 4t$, $\omega = 4$,

$$\mathbf{V}_s = 10 \angle 0^\circ \text{ V}$$

The impedance is

$$\mathbf{Z} = 5 + \frac{1}{j\omega C} = 5 + \frac{1}{j4 \times 0.1} = 5 - j2.5 \, \Omega$$



Hence the current

$$\begin{aligned}\mathbf{I} &= \frac{\mathbf{V}_s}{\mathbf{Z}} = \frac{10\angle 0^\circ}{5 - j2.5} = \frac{10(5 + j2.5)}{5^2 + 2.5^2} \\ &= 1.6 + j0.8 = 1.789\angle 26.57^\circ \text{ A}\end{aligned}$$

The voltage across the capacitor is

$$\begin{aligned}\mathbf{V} &= \mathbf{I}\mathbf{Z}_C = \frac{\mathbf{I}}{j\omega C} = \frac{1.789\angle 26.57^\circ}{j4 \times 0.1} \\ &= \frac{1.789\angle 26.57^\circ}{0.4\angle 90^\circ} = 4.47\angle -63.43^\circ \text{ V}\end{aligned}$$

$$i(t) = 1.789 \cos(4t + 26.57^\circ) \text{ A}$$

$$v(t) = 4.47 \cos(4t - 63.43^\circ) \text{ V}$$

$i(t)$ leads $v(t)$ by 90° as expected

Exercise -2

Find the input impedance of the circuit

Assume that the circuit operates at $\omega = 50$ rad/s.

Solution:

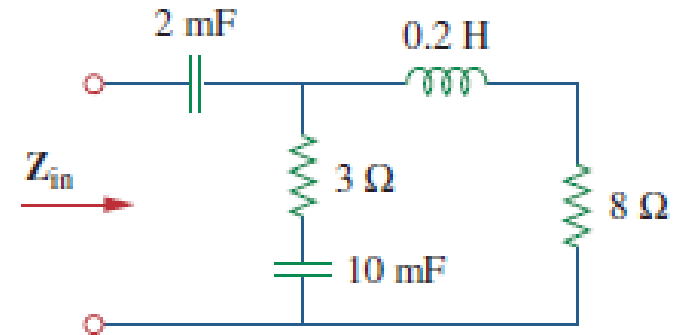
Let

Z_1 = Impedance of the 2-mF capacitor

Z_2 = Impedance of the 3- Ω resistor in series with the 10-mF capacitor

Z_3 = Impedance of the 0.2-H inductor in series with the 8- Ω resistor

$$Z_1 = \frac{1}{j\omega C} = \frac{1}{j50 \times 2 \times 10^{-3}} = -j10 \Omega$$



$$Z_2 = 3 + \frac{1}{j\omega C} = 3 + \frac{1}{j50 \times 10 \times 10^{-3}} = (3 - j2) \Omega$$

$$Z_3 = 8 + j\omega L = 8 + j50 \times 0.2 = (8 + j10) \Omega$$

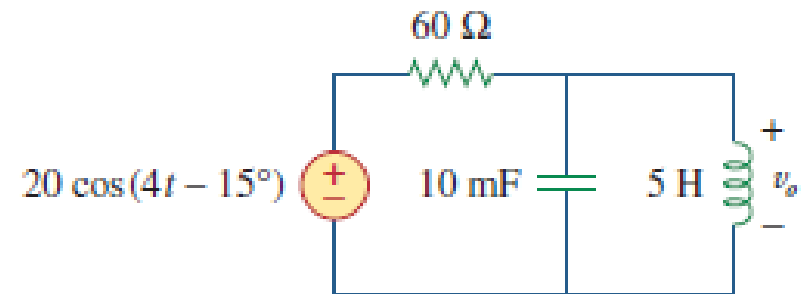
$$\begin{aligned} Z_{\text{in}} &= Z_1 + Z_2 \parallel Z_3 = -j10 + \frac{(3 - j2)(8 + j10)}{11 + j8} \\ &= -j10 + \frac{(44 + j14)(11 - j8)}{11^2 + 8^2} = -j10 + 3.22 - j1.07 \Omega \end{aligned}$$

$$Z_{\text{in}} = 3.22 - j11.07 \Omega$$

Exercise -3

Determine $v_o(t)$ in the circuit

Solution:



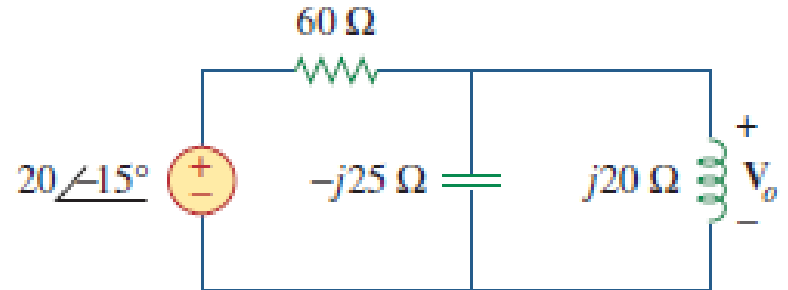
$$v_s = 20 \cos(4t - 15^\circ) \Rightarrow \mathbf{V_s = 20 \angle -15^\circ \text{ V},} \quad \omega = 4$$

$$10 \text{ mF} \Rightarrow \frac{1}{j\omega C} = \frac{1}{j4 \times 10 \times 10^{-3}} \\ = -j25 \Omega$$

$$5 \text{ H} \Rightarrow j\omega L = j4 \times 5 = j20 \Omega$$

Z_1 = Impedance of the 60- Ω resistor

Z_2 = Impedance of the parallel combination of the 10-mF capacitor and the 5-H inductor



$$Z_1 = 60\ \Omega$$

$$Z_2 = -j25 \parallel j20 = \frac{-j25 \times j20}{-j25 + j20} = j100\ \Omega$$

By the voltage-division principle,

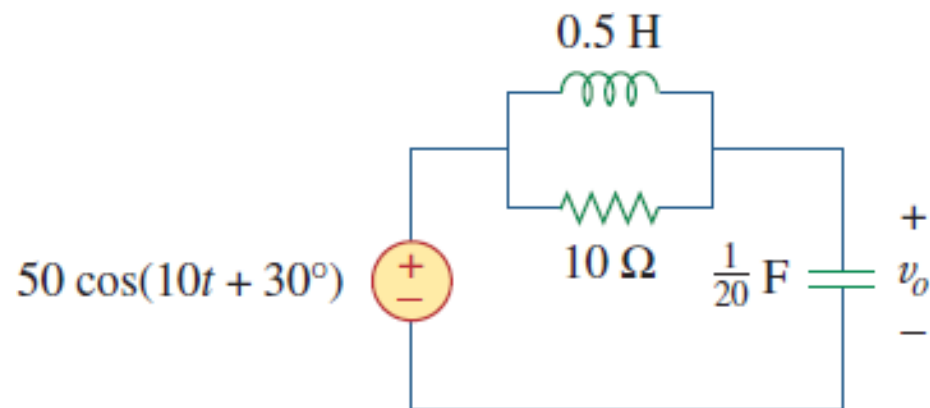
$$\begin{aligned} V_o &= \frac{Z_2}{Z_1 + Z_2} V_s = \frac{j100}{60 + j100} (20\angle -15^\circ) \\ &= (0.8575\angle 30.96^\circ) (20\angle -15^\circ) = 17.15\angle 15.96^\circ\ \text{V} \end{aligned}$$

$$v_o(t) = 17.15 \cos(4t + 15.96^\circ)\ \text{V}$$

The frequency domain equivalent of the circuit

Exercise -4

Determine $v_o(t)$ in the circuit



Answer: $v_o(t) = 35.36 \cos(10t - 105^\circ) \text{ V}$.

Thank You